Clustering-Based One-Class Classification of Remote Sensing Imagery

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2026 7th International Conference on Computer Vision, Image and Deep Learning (CVIDL) | 979-8-3195-4465-0/26/$31.00 ©2026 IEEE | DOI: 10.1109/CVIDL70130.2026.11637585

***Abstract*—One-class classification for remote sensing imagery only requires training samples from the target class, making it well-suited for scenarios where samples of other categories are scarce or difficult to acquire. Since only target-class samples are available for model training, most existing one-class classification methods rely heavily on manual parameter tuning. To tackle this limitation, this paper proposes a novel one-class classification approach based on remote sensing image clustering. Specifically, the proposed method first extracts the distribution scale of the target class from limited training samples. The derived distribution scale is then adopted to guide the clustering of remote sensing images. Finally, one-class classification is accomplished according to the obtained clustering results. We evaluate the proposed method on a multispectral remote sensing dataset and compare it with three representative approaches: one-class support vector machine, nearest neighbor classifier, and deep learning-based one-class classification. Experimental results demonstrate that the proposed clustering-driven method achieves superior classification accuracy. Furthermore, this paper proposes two improved strategies. The first is an object-oriented analysis-based method that eliminates noise interference in sample selection and clustering. The second introduces a locally adaptive clustering technique to address cross-category scale differences and intra-class multimode characteristics. Experimental results verify the superior performance of both improved schemes.**

Keywords-clustering; one-class classification; remote sensing

I. INTRODUCTION

In conventional multi-class classification, training data from two or more categories are available, and each unknown sample is assigned to one of the predefined classes. By contrast, one-class classification (OCC) is characterized by only one target class being adequately represented in the training set, whereas other classes contain no or very few labeled samples [1,2]. This inherent imbalance makes OCC more challenging than traditional multi-class classification. Even so, OCC has attracted widespread attention across numerous practical applications. It has been widely applied in anomaly detection [3] and novelty detection [4,5]. In biometric systems, spoof detection is also commonly formulated as an OCC problem [6]. Consequently, the development of advanced OCC algorithms has become a research hotspot in computer vision and machine learning. Against this background, this study focuses specifically on the one-class classification of remote sensing imagery.

Traditional remote sensing classification methods typically assign category labels to all pixels across the entire scene, which are generally regarded as multi-class classification. In practical scenarios where users are only interested in a single specific land cover category and ignore other land types, one-class classification serves as a more suitable alternative. Different from multi-class classification that requires training samples covering a complete category system, one-class classification is trained solely on samples of the target class. In the OCC framework, the category of interest to be extracted is defined as the positive class, and all remaining categories are collectively regarded as negative classes. One-class classification has been widely applied in remote sensing, including paddy rice mapping [7], urban boundary extraction [8], and building extraction [9]. Furthermore, OCC becomes the only feasible solution for situations where it is theoretically impossible to define a limited set of categories, or the non-target classes present strong heterogeneity.

This study concentrates on the one-class classification of remote sensing imagery. Over the years, a variety of one-class classification algorithms have been developed, among which kernel-based methods are the most prevalent [10]. A representative paradigm is the one-class support vector machine (OCSVM) [11], which aims to find an optimal hyperplane with the maximum margin to separate the target class from the origin in a high-dimensional feature space. These methods enable the identification of the target class using only its labeled training samples. In addition, some one-class classification methods further leverage unlabeled data to boost performance [12]. Such approaches construct binary classifiers based on positive and unlabeled samples and have exhibited promising performance in one-class classification tasks. Another kernel-based framework presented in [13] adopts a kernel sparse representation model to implement one-class classification for remote sensing images. Compared with OCSVM, nearest neighbor (NN)-based one-class classification enjoys a much simpler implementation. It determines the category of each test sample according to its distance to the k-th nearest neighbor within the target training set [14].

Over the past decade, deep learning-based object detection and recognition algorithms have become the mainstream

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research direction, achieving performance that substantially outperforms conventional machine learning methods. Accordingly, deep learning (DL) techniques have also been introduced to address the one-class classification of remote sensing imagery [15-17]. For land cover mapping applications, researchers have developed deep neural network-based approaches to implement one-class classification using remote sensing datasets [18]. In addition to algorithm design, other research aspects of one-class classification have also been extensively explored. Song et al. [19] proposed two novel sparse representation-based feature selection methods tailored for remote sensing one-class classification. Furthermore, several studies have integrated clustering techniques into the one-class classification framework. Krawczyk et al. [20] developed a method based on feature-space data clustering, where individual one-class classification models are constructed for each cluster. This method forms an ensemble of one-class classifiers and is capable of handling both one-class and multi-class classification tasks.

This paper proposes a novel one-class classification method based on remote sensing image clustering, which focuses on the target class and its distribution scale characteristics. The optimal clustering scale can be automatically determined by analyzing the inherent distribution properties of target-class training samples. Unlike most existing one-class classification approaches that rely on multiple parameters or require both positive and negative training samples, the proposed method only needs labeled samples from the target class and enjoys the advantage of fewer parameter requirements and simpler parameter configuration.

II. Methods

This study proposes a nonparametric one-class classification method based on image clustering. First, the distribution scale of the target class is quantitatively estimated from its training samples. Next, the remote sensing image is clustered under a proper scale, such that the generated clusters maintain a similar scale characteristic to the target class. Finally, one-class classification is implemented according to the clustering results. Overall, the proposed method consists of three key procedures: target class distribution scale extraction, clustering scale determination, and clustering-based one-class classification.

A. Distribution Scale of the Target Class

A key characteristic of one-class classification is that it only requires positive or unlabeled samples. Positive samples are employed to acquire statistical information and perform density function estimation for the target class. Common density estimation strategies include parametric and non-parametric methods. In the one-class classification scenario, the distribution of negative classes is generally unknown. Therefore, it is essential to adequately characterize the target class to achieve effective separation from other categories. The distribution characteristics of the target class are typically described by statistical indicators derived from training samples, such as mean vectors and covariance matrices. To further improve classification performance, more inherent distribution properties of target samples should be extracted and exploited, among which the class distribution scale is particularly important. In this paper, distribution scale refers to the distribution range and density characteristics of a class within the feature space. As illustrated in Fig. 1, the two presented classes possess distinct distribution scales: the left class exhibits a large distribution scale, while the right one has a relatively small scale.

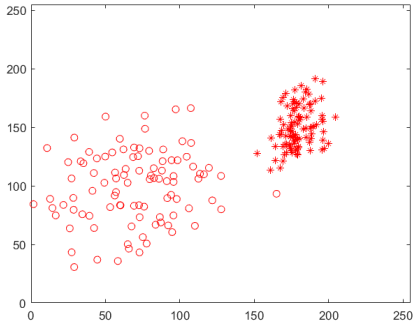


Figure 1. Different class distribution scales

Land cover classes in remote sensing images exhibit complex and diverse distributions in the feature space, with distinct differences in distribution scales. For instance, in multispectral imagery, water bodies typically present a compact small-scale distribution with a narrow and dense feature space range. By comparison, woodland usually occupies a substantially larger feature space region. Accordingly, the distribution scale of the target class is an indispensable consideration for one-class classification tasks.

One-class classification is essentially a supervised learning paradigm trained solely on target-class samples. Meanwhile, each land cover class in remote sensing datasets presents unique intrinsic distribution patterns in the feature space, including density, shape, and spatial extent. Accordingly, the performance of one-class classification relies not only on discriminative features derived from target training samples but also on the overall distribution characteristics of all classes within the original dataset. As an effective unsupervised data analysis tool, clustering is capable of mining inherent class distribution patterns hidden in datasets. By integrating clustering into the one-class classification framework, supervised classification can be implemented collaboratively by leveraging both the global distribution characteristics of all classes and the local feature information extracted from target-class training samples.

Each target class possesses a unique inherent distribution scale. Statistically, such scale can be characterized by the intra-class scatter matrix, which is essentially derived from the pairwise distances among samples. To simplify the computation, this paper presents a scale measurement approach based on the average Euclidean distance between sample pairs. Given a dataset with L samples, all samples are randomly partitioned into $L/2$ groups. Let Di denote the Euclidean distance between the two samples within the i-th group. The average pairwise distance D is then defined as follows:

$$D = \text{mean}\,(\,Di\,). \tag{1}$$

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B. Selection of Scale in Clustering

The proposed one-class classification method is built upon unsupervised data clustering. Clustering partitions data points into distinct groups according to their feature similarity, where samples within the same cluster exhibit high consistency and present a stable distribution scale in the feature space. To achieve reliable classification, the clusters corresponding to the target class should possess a distribution scale analogous to that of target training samples. Accordingly, the selection of an appropriate clustering scale is crucial for effective remote sensing image clustering.

The clustering assignment of a data point depends on both its position in the feature space and the distribution scale of each cluster. Cluster scale refers to the extent of a cluster's distribution range within the feature space and is governed by the scale parameter of the clustering algorithm. In different clustering algorithms, the scale parameters are different. For example, the scale parameter in k-means clustering method is the given number of clusters, while in mean-shift based clustering it is the bandwidth [21]. Clustering results derived from different scale parameters differ in the number, spatial distribution, and density of the resulting clusters. Taking cluster quantity as an example, the scale parameter is closely correlated with clustering performance: a larger scale yields fewer clusters, while a smaller scale generates more clusters. Conversely, a greater number of output clusters implies the adoption of a small scale parameter, and the inverse also holds true.

Although a dataset possesses an inherent class structure in the feature space, it can be interpreted and analyzed at multiple scales. A single intrinsic class can be treated as one integral cluster at a large analysis scale, while it may be partitioned into multiple sub-clusters at a small scale. Such classes present a composite distribution pattern consisting of multiple internal sub-patterns. When adopting a smaller scale, a well-performed clustering algorithm tends to generate more compact clusters with reduced intra-cluster distances. In other words, a greater number of clusters corresponds to a smaller distribution scale. In practice, different classes often exhibit partial overlaps even when their cluster centers are clearly distinguishable, which is a common characteristic of remote sensing image data. Given the inherent complexity of remote sensing datasets, it is therefore essential to fully exploit the inherent distribution characteristics of the data itself.

The scale parameter exhibits inconsistent functional mechanisms across different clustering algorithms, making it impossible to adopt a unified setting. Even for a single clustering algorithm, different scale parameters can produce markedly different clustering outcomes. Accordingly, the optimal scale can be inferred directly from the clustering results, which further enables the definition of a generalized clustering scale applicable to diverse clustering algorithms.

Since the clustering scale is determined based on clustering results, tentative clustering with different scale parameters is required to obtain a result with an appropriate scale. For each clustering result, the mean pairwise distance between data points within each cluster is first calculated as the individual cluster scale D_ci. The average of all D_ci values is then taken as the overall scale D_c of the clustering result. Among multiple clustering outcomes generated with different parameters, the one whose D_c is closest to the training sample scale D_s is selected. One-class classification is subsequently implemented based on this optimal clustering result. In this manner, the clustering scale selection inherently incorporates the threshold determination for one-class classification. Consequently, the one-class classification method proposed in this study is free of manual parameter configuration.

C. Clustering-based One-Class Classification

The clustering-based one-class classification (CBOCC) method proposed in this study first performs cluster labeling on clustering outputs. Given that clusters of the dataset are generated via a clustering algorithm, a cluster is labeled positive once it contains any training samples from the target class. If the target-class training samples are distributed across multiple clusters, all these corresponding clusters are assigned positive labels. Accordingly, multiple positive clusters can be yielded rather than a single one. Subsequently, the distribution scale D_c is computed for all positive clusters derived from each clustering result. Finally, the clustering result with optimal D_c is selected, and samples within the positive clusters of this optimal result are assigned to the target class. The detailed implementation procedure of the proposed one-class classification framework is elaborated in the following.

Suppose there are n training samples of target class from the dataset N, and the training sample set S,

$$S = \{ S_1, S_2, \dots , S_n\} \tag{2}$$

Clustering the dataset N to generate m clusters, each cluster is represented by C_i, then

$$N = \{ C_1, C_2, \dots , C_m\} \tag{3}$$

For a cluster C_i, if it intersects with points in the training sample set S of the target class, C_i is a positive cluster, otherwise it is a negative cluster.

$$C_i \in \begin{cases} \text{positive} & C_i \cap S \neq \emptyset \\ \text{negative} & C_i \cap S = \emptyset \end{cases} \tag{4}$$

If a sample X_j is in the positive cluster, it is classified as positive class; otherwise it is classified as negative class.

$$\mathrm{X}_j \in \begin{cases} \text{positive} & \mathrm{X}_j \in C_i \text{ and } C_i \in \text{positive} \\ \text{negative} & \mathrm{X}_j \in C_i \text{ and } C_i \in \text{negative} \end{cases} \tag{5}$$

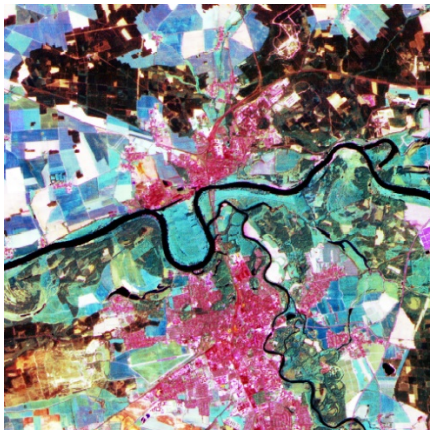
The clustering-based one-class classification method fully exploits the intrinsic distribution structure of datasets. For certain remote sensing data, such as Thematic Mapper (TM) images, the visible bands exhibit strong inter-band correlation, which physically causes spectral overlap among different land-cover classes. These classes present distinct data distributions with varying overlap degrees. Accordingly, it is essential to extract distribution characteristics for reliable target class characterization in one-class classification. In the proposed CBOCC method, the distribution scale is automatically learned from target-class training samples, thereby addressing the threshold selection challenge inherent to one-class classification.

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III. Experimental Results

A. Experimental Data

A subset of a Landsat Thematic Mapper (TM) scene was adopted for experimental validation. Fig. 2(a) presents the original 498 × 499 multispectral image displayed in an RGB composite format. The digital numbers (DNs) of all bands in the experimental image range from 0 to 255; thus, no data rescaling was performed. Fig. 2(b) illustrates the reference classification map utilized for quantitative accuracy evaluation. This ground-truth map was generated via manual visual interpretation, where red denotes impervious surfaces, blue represents water bodies, yellow indicates grassland, and green corresponds to woodland. Referencing the ground-truth map, 5% of pixels belonging to the target class were randomly sampled from Fig. 2(a) for model training, while the remaining 95% were retained for testing.



(a)



(**b**)

Figure 2. Experimental data: (**a**) TM image; (**b**) reference classification.

B. Implementation Detail

Taking impervious surfaces as the research example, target-class training samples were selected from the remote sensing image, as visualized in Fig. 2. The average pairwise distance between sample points was calculated from the training set and defined as the target class scale D_s. Subsequently, the mean-shift clustering algorithm [21] was employed to segment the TM image. The clustering scale D_c was computed based on the positive clusters derived from the clustering outputs. The detailed selection criteria for the optimal clustering scale are specified as follows.

- 1) Calculate the reference scale D_s of the target class from labeled training samples, and determine a permissible scale tolerance threshold ε.
- 2) Initialize the initial bandwidth B_0 and a fixed iteration step ΔB for the mean-shift algorithm.
- 3) Cluster the experimental remote sensing image using mean-shift with the current bandwidth parameter B.
- 4) Calculate the clustering scale D_c from the clustering result.
- 5) Adjust the bandwidth adaptively: decrease B if $D_c > D_s$, and increase B if $D_c < D_s$ with the fixed step ΔB.
- 6) Iteratively repeat Steps 3–5 until the absolute scale deviation satisfies $|D_c - D_s| \leq \varepsilon$ or the iteration reaches stable convergence.
- 7) Select the optimal bandwidth corresponding to the D_c with the minimum deviation from D_s as the final mean-shift bandwidth.

The mean-shift algorithm adopts the Epanechnikov kernel, where the window radius serves as the bandwidth parameter. This bandwidth parameter B governs the clustering scale and varies with specific algorithms and feature spaces. In this study, all mean-shift clustering experiments were implemented in the RGB feature space. The initial bandwidth B_0 and the iteration step ΔB were set to 50 and 10 respectively in Step 2. Specifically, if the obtained clustering scale D_c exceeds D_s, the bandwidth is reduced to 40 for the next iteration, and this adjustment continues iteratively. The iteration process terminates when the termination condition in Step 6 is satisfied, where the scale tolerance threshold ε was set to one percent of D_s. One-class classification is subsequently implemented on the optimized clustering result. Specifically, any cluster containing target-class training samples is defined as a positive cluster, and all pixels within such clusters are classified as positive samples.

C. One-class Classification Results

To demonstrate the validity of the proposed clustering scale selection strategy, impervious surfaces were selected as the experimental example for one-class classification under different clustering outputs. Three quantitative metrics, namely producer's accuracy, user's accuracy, and overall accuracy, were adopted to quantitatively assess the classification performance. The visual classification results are displayed in Fig. 3, while the corresponding quantitative accuracy metrics are listed in Table 1, where the bandwidth denotes the scale parameter of the mean-shift algorithm. In Table 1, the column "clusters/target" indicates the total number of generated clusters and the quantity of target-containing clusters. Notably, the Euclidean distance was calculated without square-root operation to reduce computational consumption. In the experiment, the calculated D_s of impervious surface training samples was 465.

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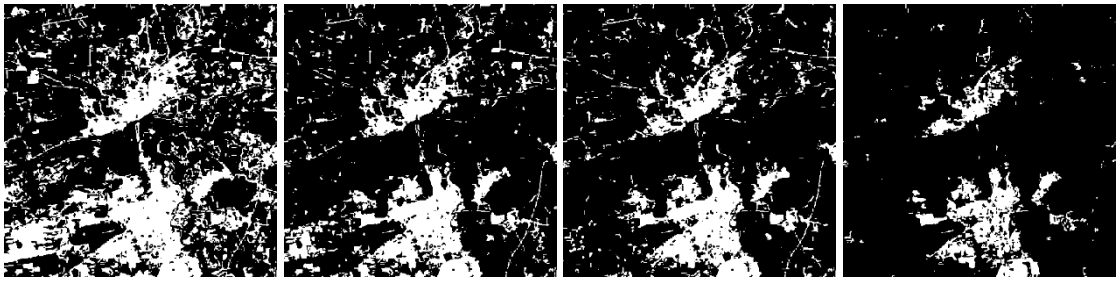


Figure 3. One-class classification of impervious with bandwidth from left to right: 50, 40, 30, 20.

TABLE I. CLASSIFICATION ACCURACY OF IMPERVIOUS

bandwidth	clusters/target	D_c	PA	UA	OA
50	5/3	1321	0.865	0.452	0.806
40	8/4	1177	0.807	0.669	0.903
30	21/8	650	0.806	0.737	0.921
20	132/15	186	0.556	0.94	0.922

As summarized in Table 1, varying bandwidth parameters yield clustering results with different scales. Specifically, a larger bandwidth corresponds to a broader clustering scale. Such scale differences further induce variations in one-class classification accuracy. Among all experimental groups, the clustering scale of 650 presents the minimum deviation from the target scale *D_s*, and the corresponding classification achieves the highest producer's accuracy, user's accuracy, and overall accuracy. Hence, 650 is determined as the optimal clustering scale for impervious surface extraction. As illustrated in Fig. 3, the coverage of individual clusters gradually shrinks with the decrease in clustering scale, which correspondingly narrows the distribution range of the identified target areas. Therefore, an optimal clustering scale exists to maximize the one-class classification performance for the target land-cover class.

Similarly, the one-class classification was conducted for the remaining land-cover classes within the experimental image, and the corresponding classification results are presented in the first row of Fig. 4. For comparative analysis, the proposed method was benchmarked against three conventional one-class classification algorithms, namely OCSVM, nearest neighbor (NN)-based one-class classification, and a deep learning (DL)-based approach. For the OCSVM model, the radial basis function (RBF) kernel was adopted, and all parameters were optimized via 10-fold cross-validation. The NN-based method employs a target-class whitening transformation [22]; following the transformation, pixels with distances to the target class center below a predefined threshold are categorized as target samples. The threshold of the NN classifier was empirically set to three times the standard deviation of the target class mean. Furthermore, a convolutional neural network (CNN)-based deep learning model was trained on the training set derived from the TM remote sensing image. Supported by ENVI Deep Learning [23], the label raster was constructed based on target-class training samples. During model training, the number of input bands and classes were configured to 3 and 1, respectively, while other hyperparameters remained as default. The one-class classification outputs of all comparative methods are visualized in Fig. 4.

Table 2 quantitatively summarizes the classification accuracy achieved by different comparative methods. The proposed CBOCC method achieved an overall accuracy comparable to the OCSVM approach. Notably, OCSVM relies on cross-validation for parameter optimization, whereas the presented CBOCC method autonomously extracts the clustering scale from training samples without additional manual parameter configuration. Furthermore, OCSVM requires both positive and negative samples for cross-validation during model training, while CBOCC only demands positive samples, which better conforms to the application characteristics of one-class classification. For the NN-based classifier, classification performance is highly sensitive to threshold selection, and the empirical threshold (i.e., three times the standard deviation of the class mean) lacks universal applicability across diverse scenarios. Moreover, restricted by the limited training samples, the deep learning method failed to yield prominent accuracy improvement despite its relatively complicated training workflow. Even without manual parameter tuning, the proposed CBOCC method achieved competitive classification performance against the compared algorithms.

TABLE II. ACCURACY COMPARISON BETWEEN VARIOUS METHODS

classes	PA				UA				OA			
	CBOCC	*OCSVM*	*NN*	*DL*	*CBOCC*	*OCSVM*	*NN*	*DL*	*CBOCC*	*OCSVM*	*NN*	*DL*
impervious	0.806	0.774	0.758	0.847	0.737	0.677	0.729	0.687	0.921	0.903	0.915	0.912
woodland	0.851	0.736	0.726	0.886	0.804	0.923	0.851	0.825	0.865	0.877	0.858	0.886
grassland	0.718	0.723	0.830	0.783	0.903	0.870	0.820	0.922	0.845	0.834	0.859	0.878
water	0.700	0.729	0.756	0.790	0.943	0.978	0.964	0.810	0.991	0.992	0.992	0.989

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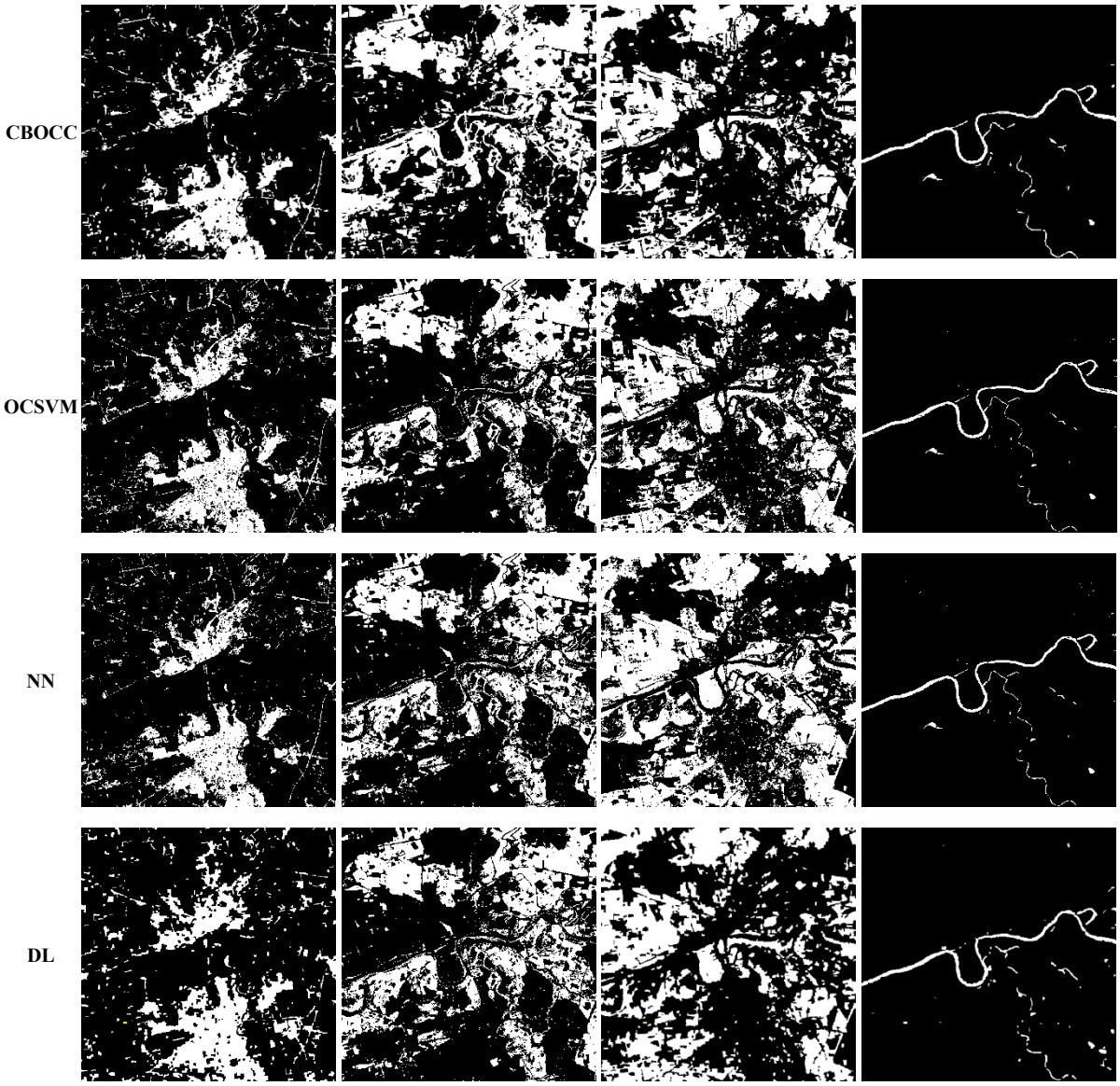


Figure 4. Comparison of one-class classifications, from left to right: impervious, woodland, grassland, water bodies.

D. Object-oriented analysis

The proposed method assigns category labels to entire clusters relying on the contained training samples, making the classification results sensitive to noise within training samples. To alleviate such sensitivity, it is essential to eliminate noise interference from the training sample set. Accordingly, an object-oriented image analysis strategy is adopted in this paper. First, the remote sensing image is segmented to construct homogeneous image objects, and the proposed CBOCC method is then implemented on the basis of these image objects. Numerous image segmentation methods exist and a region merging-based method is used to segment TM images in this study. The segmentation result is illustrated in Fig. 5. Subsequently, the CBOCC algorithm is applied to the segmented image objects, and the obtained one-class classification result is presented in Fig. 6. Quantitative accuracy evaluation for each category is listed in Table 3.

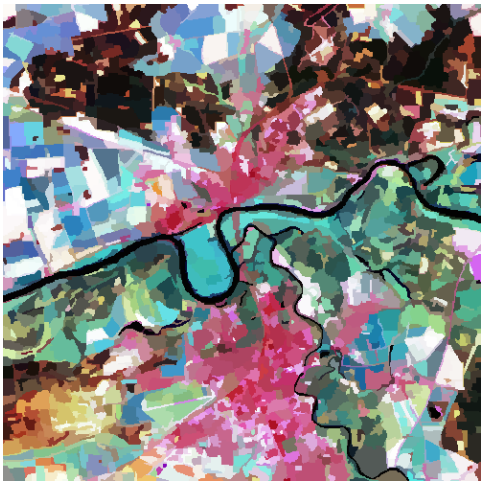


Figure 5. Segmented image.

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E. Adaptive mean-shift clustering

As shown in Fig. 4, water bodies are misclassified into the woodland category. This phenomenon occurs because several water body and woodland samples are grouped into the same clusters during the clustering procedure. Overall, water bodies present a small distribution scale in the feature space, whereas woodland occupies a much larger scale. When clustering is performed based on the large scale of woodland, water body samples that are spectrally adjacent to woodland samples in the feature space tend to be merged into identical clusters. To address this problem, a clustering strategy with locally adaptive scale is introduced. This approach adopts varying clustering scales for different local regions in the feature space. Specifically, a small scale is applied to local areas containing water body samples, while a relatively large scale is adopted for regions dominated by woodland samples. This scheme effectively avoids the clustering defects caused by a fixed unified scale. In this study, an adaptive mean-shift clustering algorithm is employed, in which the clustering bandwidth is automatically determined according to the local feature density around each sample [24]. The advantage of the adaptive mean-shift method lies in its capability to accurately partition clusters with inconsistent feature space densities, thereby achieving optimal cluster segmentation. One-class classification is further implemented based on the well-partitioned clustering results. In the experiment, adaptive mean-shift clustering is applied to the segmented image, and the corresponding one-class classification result is displayed in Fig. 7.

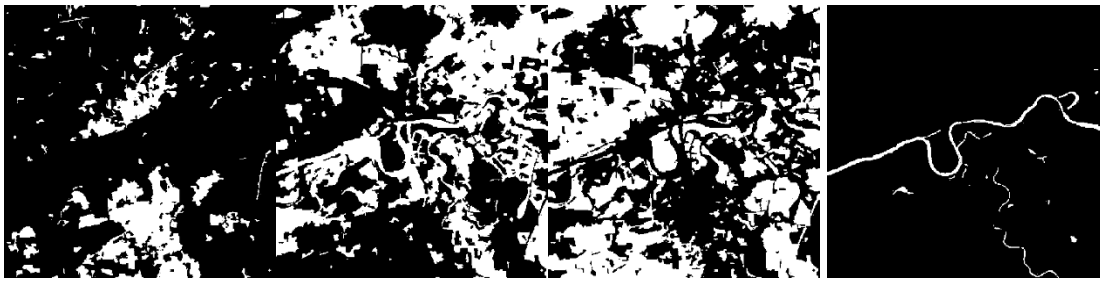


Figure 6. One-class classification based on segmented image, from left to right: impervious, woodland, grassland, water bodies.

TABLE III. CLASSIFICATION ACCURACY OF OPTIMIZATIONS

classes	PA			UA			OA		
	CBOCC	***Obj***	***Ada***	***CBOCC***	***Obj***	***Ada***	***CBOCC***	***Obj***	***Ada***
impervious	0.806	0.716	0.904	0.737	0.837	0.789	0.921	0.931	0.945
woodland	0.851	0.869	0.806	0.804	0.835	0.895	0.865	0.885	0.891
grassland	0.718	0.884	0.938	0.903	0.867	0.839	0.845	0.892	0.896
water	0.700	0.762	0.779	0.943	0.938	0.933	0.991	0.992	0.992

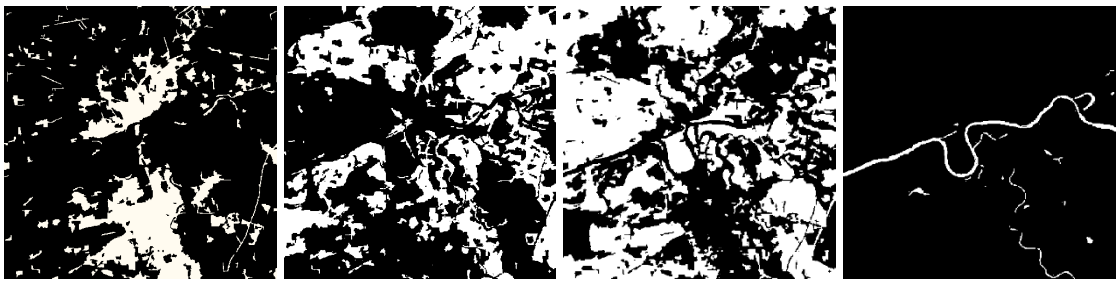


Figure 7. One-class classification based on adaptive clustering, from left to right: impervious, woodland, grassland, water bodies.

Table 3 quantitatively summarizes the classification accuracy of the original CBOCC method and its two optimized variants. In the table, “Obj” denotes the CBOCC method based on object-oriented analysis, while “Ada” represents the version integrated with adaptive clustering. It can be observed that both optimization strategies effectively improve the overall one-class classification accuracy. In particular, the adaptive clustering variant achieves a significant accuracy improvement for the woodland category, which enables reliable separation between woodland and water body features, as illustrated in Fig. 7.

IV. CONCLUSIONS

This study presents a clustering-driven one-class classification framework for remote sensing imagery. The image was firstly segmented using the mean-shift algorithm, and the clustering outputs were subsequently employed for one-class classification. Comparative experiments were conducted on a TM remote sensing subset against OCSVM, NN-based classifier, and DL-based method. The experimental results demonstrate that the proposed CBOCC achieves competitive classification accuracy compared with the benchmark methods while possessing the distinct advantage of simple parameter configuration. In addition, this paper proposes two improved schemes. The first is an object-oriented

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analysis-based method, which avoids the interference of noise during sample selection and clustering. The second improvement adopts a locally adaptive technique in the clustering process, addressing the problems that different categories in images possess distinct distribution scales and a single category contains multiple patterns. Experimental results verify that both improved schemes achieve satisfactory performance. However, there are still certain inherent limitations. Remote sensing images contain abundant multispectral information, such as NIR and SWIR bands, as well as vegetation and surface feature indices including NDVI, which can provide more discriminative feature representations for image clustering and target extraction. Although this study focuses on verifying the superiority and applicability of the proposed one-class classification framework under a unified RGB feature condition to maintain experimental consistency and controllability, further exploring the adaptability and performance differences of the proposed method in multispectral band combinations and spectral index feature spaces is still worthy of in-depth investigation.

ACKNOWLEDGMENT

This work was supported by the National Natural Science Foundation of China (U1904119).

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